KIT-SIM

Daylight report: Harbour Quarter, Block 4

Run complete · Run R-0724 · 24 Oct 2026, 14:32 · kit-sim 0.4.2 · kit-ui 0.1.0

MEAN DA	SDA 300/50	ZONES PASSING	GRID POINTS
62%	66%	4 / 6	6 508
Area-weighted	Target 55%	Core fails	1 627 m² at 0.5 m

Summary

Block 4 reaches a spatial daylight autonomy of 66% across its 1 627 m² of floor area, above the 55% target. Four of six zones pass. The core falls well short at 22% and needs either a deeper light well or a change of use; the courtyard wing sits just under the target and is flagged for review.

Inputs

Site and model

PARAMETER	VALUE
Location	Hamburg, 53.55° N 9.99° E
Weather file	Hamburg TMYx, WMO 101470
Model	block4_v12.glb, 12.4 MB
Floors	4, ground to third
Glazing	Visible transmittance 0.62

Simulation

PARAMETER	VALUE
Analysis grid	0.5 m, 0.85 m above floor
Occupied hours	08:00-18:00, Mon-Fri
DA threshold	300 lx
Ambient bounces	5
Sky model	Perez all-weather, hourly

Results

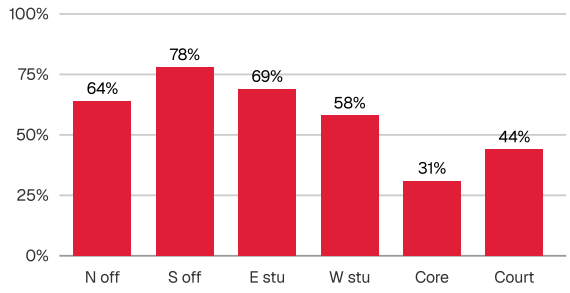


Figure 1. Mean daylight autonomy per zone: share of occupied hours at or above 300 lx.

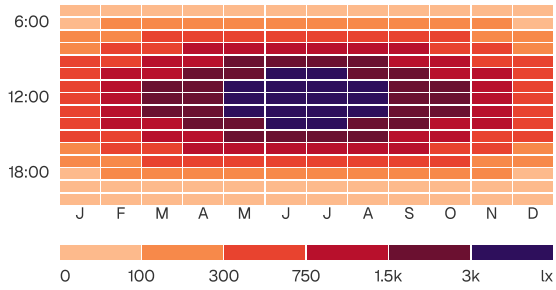


Figure 2. Mean illuminance over the block by hour and month.

Table 1. Results per zone. sDA passes at 55%; 40-55% is flagged for review.

ZONE	AREA (m²)	MEAN DA (%)	SDA 300/50 (%)	RESULT
North office	412	64	71	Pass
South office	398	78	88	Pass
East studio	236	69	76	Pass
West studio	241	58	61	Pass
Core	188	31	22	Fail
Courtyard	152	44	47	Review

Method and assumptions

Illuminance is computed hourly for every grid point with a Perez sky from the weather file. Daylight autonomy (DA) is the share of occupied hours a point reaches 300 lx on daylight alone; spatial daylight autonomy (sDA 300/50) is the share of a zone's points with a DA of at least 50%.

Blinds are not modelled and neighbouring buildings are taken from the site model as delivered. Results for the core assume the light well finishes at a reflectance of 0.7.

Appendix

Table A1. Mean DA per zone and floor (%). Every floor holds the same grid points per zone.

ZONE	POINTS PER FLOOR	FLOOR 0	FLOOR 1	FLOOR 2	FLOOR 3
North office	412	58	62	66	70
South office	398	72	76	80	84
East studio	236	63	67	71	75
West studio	241	52	56	60	64
Core	188	25	29	33	37
Courtyard	152	38	42	46	50

Table A2. Provenance

ITEM	VALUE
Run	R-0724, started 24 Oct 2026 14:26, finished 14:32 (6 min 12 s)
Application	kit-sim 0.4.2 (Daylight), kit-ui 0.1.0
Engine	Radiance 5.4, rtrace -ab 5 -ad 2048
Model hash	`sha256:4f1c9e…a20b`